

KULASEKARAN M

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PROFESSIONAL SUMMARY

Embedded Firmware Engineer with expertise in STM32, ESP32, Raspberry Pi, and RTOS development. Proven experience implementing real-time computer vision systems, IoT protocols (MQTT, HTTP/HTTPS), and AWS cloud integration. Proficient in C, Python, hardware debugging, and communication protocols (SPI, I2C, UART). Strong background in ARM architecture, embedded systems optimization, and machine learning integration for IoT applications.

TECHNICAL SKILLS

Programming Languages: C, Python, Assembly Language, VHDL
Embedded Systems: ESP32, Raspberry Pi, Jetson AGX Xavier, ARM Cortex-M, FPGA
RTOS & Protocols: FreeRTOS, SPI, I2C, UART, MQTT, HTTP/HTTPS, MAVLink
Development Tools: STM32CubeIDE, Keil µVision, GDB, Git, Oscilloscope, Logic Analyzer
Cloud & AI/ML: AWS (S3, Lambda, IoT Core), YOLOv5, OpenCV, Computer Vision, Machine Learning
Hardware Design: PCB Design (KiCad), Cadence, Circuit Debugging, Power Management

PROFESSIONAL EXPERIENCE

R&D Embedded Engineer

BB IoT Solutions Pvt Ltd

June 2024 – July 2025

Vellore, Tamil Nadu

- Developed real-time object recognition system integrating computer vision and RFID technology, achieving **68% to 92% accuracy YOLO + RFID fusion** for industrial IoT applications
- Maintained **99.5% uptime** over 6 months in production handling 15K+ MQTT messages/day in production environment
- Collaborated with cross-functional teams to optimize firmware for memory efficiency and power consumption in resource-constrained embedded devices

PUBLICATIONS

Advanced Edge-Cloud Offloading System for Industrial Quality Control

Research Paper – Real-Time Embedded Systems

Yet to be Published

- Designed RTOS-based embedded system with intelligent edge-cloud workload distribution for post-manufacturing chip defect detection under CPU, thermal, and power constraints
- Achieved **30% reduction in edge CPU utilization** through dynamic cloud offloading while maintaining real-time performance requirements using MQTT/HTTP communication

TECHNICAL PROJECTS

Embedded mmWave Radar Signal Acquisition and Motion Detection System using ESP32-S3 | SPI, FMCW Radar

- Interfaced Infineon BGT60TR13C 60GHz FMCW radar sensor with ESP32-S3 using high-speed SPI for real-time data acquisition
- Configured radar registers and implemented firmware pipeline to read raw ADC data from on-chip FIFO
- Developed lightweight signal processing (energy-based detection) to enable motion and presence detection under resource constraints
- Optimized data handling and communication to manage high-throughput radar streams while maintaining real-time system responsiveness

Edge AI Enabled Autonomous Disaster Response Drone With Multi-sensor Perception |

Published

- Built autonomous disaster response drone with S500 quadcopter frame, Pixhawk flight controller, and multi-sensor payload (RGB/thermal cameras, gas sensors MQ2/MQ7)
- Integrated sensor suite using I2C and SPI protocols with real-time telemetry via MAVLink communication protocol
- Implemented YOLOv5 object detection on Raspberry Pi for real-time target identification, achieving **85% accuracy** in disaster simulation scenarios

Secure Embedded Linux Multi-Sensor Surveillance System with Kernel Drivers and Firmware Integrity Protection | I2C, V4L2

- Designed and implemented custom Linux kernel I2C device drivers for BME280 and APDS9960 sensors with interrupt-driven detection and character device interfaces
- Developed real-time video/audio capture system using V4L2 and ALSA with memory-mapped buffer management, achieving sub-50ms event-triggered recording latency
- Implemented firmware integrity verification using periodic cryptographic hashing to detect tampering and prevent malicious firmware modification
- Designed lightweight security mechanism to restrict unauthorized updates and ensure reliable operation under potential physical and software-level attacks.

Hardware-Accelerated Multi-Stage Appliance Control System Using FPGA and VHDL | VHDL, FSM, Xilinx Vivado, FPGA

- Designed and implemented a Finite State Machine (FSM)-based control system for automating washing machine operations including water fill, wash, rinse, drain, and spin cycles
- Developed VHDL behavioral code for state transitions, timing control, and output logic using synchronous design principles
- Achieved reliable and efficient hardware-level automation, improving flexibility and performance over traditional microcontroller-based designs

EDUCATION

Vellore Institute of Technology

Master of Technology (M.Tech) in Embedded Systems; CGPA: 8.19/10

Vellore, Tamil Nadu
Pursuing

Vellore Institute of Technology

Bachelor of Technology (B.Tech) in Electronics and Communication Engineering; CGPA: 7.88/10

chennai, Tamil Nadu
2020 – 2024

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

AWS Certification – [AWS Certified Cloud Practitioner](#) (2024)

BerriBot Pvt Ltd [Internship] – [AI Testing Engineer](#) (May – July 2024)

PCB Design Using KiCad – [KiCad Workshop Certification](#)

Design of High-Speed VLSI Circuits Using Cadence – [VLSI Cadence](#)

24hrs Industrial Conclave Hack-a-thon – [Winner](#)

8-Hours NetSim Hackathon 2.0 – [Participant](#)

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

IEEE Student Chapter, VIT

Student Coordinator

- * Coordinated 12+ technical sessions and guest lectures.
- * Organized embedded systems workshop with 150+ participants.
- * Increased membership engagement by 35%.

2020 – 2024
chennai, Tamil Nadu

Wayspire EdTech

College Ambassador

- * Acted as liaison between organization and student body, increasing internship participation by **20%**

2023 – 2024